AWS GPU Quota — Analysis, Plan A & Plan B

LocalAI TV · Architecture Team · 19 May 2026

For the Founder: Koneti Mohan Reddy (Managing Director) **cc:** Nagarjuna Reddy (CTO), Sheikh Sameer & Gyana Rajnan (AI Pipeline) **Re:** AWS Service Quota — "Running On-Demand G and VT instances", 8 vCPU, ap-south-1 (Case 177892985400464) — rejection of 16 May, reopen of 18 May
Status: Advisory analysis. The Phase-0 gate is unaffected and remains held; this is about *where* the Phase-0 GPU validation runs, not about lifting the gate.

TL;DR — plain language for the Founder

Do not let one scarce AWS approval hold the whole project hostage. Here is the key insight:

- Your **Phase-0 GPU need is tiny** — *one* small GPU machine, 4–6 hours a day, for testing. That is a very small ask.
- AWS is slow to give **GPU** quota to **brand-new accounts** with **no spending history**, especially in **Mumbai** (GPU capacity is tight there). The 31-minute rejection on 16 May was an **automatic/template denial**, not a real review — this is normal and expected for a new account.
- Reopening with a detailed use case (18 May) was **exactly the right move** and genuinely improves the chances — but approval is **not guaranteed and may take days to weeks**, possibly only a partial grant.

So the smart plan is two tracks in parallel:

- **Plan A — push AWS properly:** upgrade to AWS Business Support (the single biggest lever), let your already-live AWS storage build a billing history, and re-request with that history. Also request the *Spot* GPU quota and a *different region* — these are often approved when Mumbai On-Demand is not.
- **Plan B — don't wait idle:** the Phase-0 GPU test can run on a **specialist GPU cloud (RunPod / Lambda Labs)** or an **Indian GPU cloud (E2E Networks)** *this week*, while your storage, database and CDN **stay on AWS exactly as already built**. The test reads/writes the same AWS S3 — at test volume the cross-cloud cost is a few dollars, nothing more.

Bottom line: You can begin the Phase-0 GPU validation within days regardless of AWS, by renting a single GPU elsewhere while keeping everything else on AWS. We should **not** redesign the whole platform for "multi-cloud" — just use a temporary hybrid to unblock testing, keep the GPU job portable, and consolidate once AWS quota lands.

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1. What the three screenshots actually show (the timeline)

When	What happened	Reading
16 May, 16:41	AWS: "requires collaboration with our internal teams … I will notify you."	Standard acknowledgement; routed to internal capacity team.
16 May, 17:12 (31 min later)	AWS: **"unable to approve … service quotas help you gradually ramp up … avoid large bills from sudden spikes … reopen with a detailed use case and we'll re-assess."**	This is the **standard automated/first-line denial** for a GPU quota from a young account. The 31-minute turnaround confirms it was **not** a considered review. The "reopen with detail" line is a **genuine, invited appeal path**.
18 May, 13:00	Founder reopened with a strong, detailed use case (instance, hours, budget cap, isolated account, timeline).	Correct response — this is precisely what AWS asked for and materially improves the odds.
19 May (today)	Awaiting AWS response to the reopen.	Expect days, not hours. May involve more questions.

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2. Q1 — Why is AWS not approving? (most likely, ranked)

1. **New account, no billing/usage history (biggest reason).** The isolated account `689186650531` has near-zero spend. AWS's own words — *"gradually ramp up activity"* — literally mean *"you have no track record yet."* Going from the default (often **0**) straight to **8 vCPU of GPU** is a "from-zero" jump AWS scrutinises hard.
2. **GPU capacity is constrained in ap-south-1 (Mumbai).** G/VT (NVIDIA T4) instances are scarce in Mumbai; AWS is conservative there even for mature accounts.
3. **First-line automation.** The 31-minute denial was templated, not human-assessed. Real assessment happens on the appeal — which is now in progress.
4. **No paid AWS Support plan.** Quota requests from **Basic Support** (free) are deprioritised. Business Support requests move far faster and more favourably.
5. **The "scaling to 24/7 in Phase 1" phrase in the request can backfire.** It signals exactly the "sudden large bill" risk AWS's denial text cites. (See the tactical fix in §4.)

These are normal new-account headwinds, **not** a sign anything is wrong with the project or the request.

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3. Q2 — Chances of approval after reopening?

Honest assessment: moderate, and improved — but not guaranteed, and not fast.

- The reopen is strong (clear bounded use case, $364.70 cap, tagged & isolated account, specific instance). That genuinely helps.
- The two structural headwinds remain: new-account history and Mumbai GPU scarcity.
- **Realistic outcomes:** (a) approved after one or more rounds over several days–weeks; (b) **partial** grant (e.g. 4 vCPU = one g4dn.xlarge, not 8); (c) further questions before any grant; (d) continued deferral until the account shows spend history.
- **Plan the timeline as if approval is uncertain.** Do not gate the project schedule on it (hence Plan B).

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4. Q3 — Will AWS ask for more, and what helps most?

Likely follow-ups: **billing/payment history & real spend**, business/identity verification (new India entity), a request to **ramp gradually** (smaller workloads first), or more architecture detail (already well covered).

Highest-leverage actions to convert the appeal:

1. **Upgrade to AWS Business Support (~$100/mo or 10% of spend).** Single most effective lever for a new account — quota appeals are handled by a faster, more empowered queue. Recommended regardless of Plan A/B.
2. **Let the live AWS resources accrue spend.** S3 + CloudFront + RDS are already built — even modest real billing over 1–3 weeks materially strengthens the re-request. Attach the billing history to the case.
3. **Tactical wording fix in AWS communications:** emphasise the **bounded Phase-0** (one g4dn.xlarge, 4–6 hrs/day, hard $364.70 alert, isolated account). **Soften the "24/7 Phase-1" language** — frame Phase-1 as *"gradual, post-validation, requested incrementally as usage grows."* This aligns with AWS's stated "gradual ramp" philosophy instead of triggering its "sudden spike" reflex.
4. **Add a payment method with history / a small prepayment** if available; ask if an **AWS account manager / Activate (startup) credits** apply to the entity.

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5. Plan A — Maximise AWS approval (do these now)

#	Action	Owner
A1	Upgrade the Phase-0 account to **AWS Business Support**	CTO
A2	Keep the reopened case open; respond fast to any AWS question; attach billing history as it accrues	AI team
A3	Re-request with the **softened Phase-0-bounded wording** (§4.3)	AI team
A4	Also file a **Spot** quota: *"All G and VT Spot Instance Requests"* — often granted when On-Demand is denied, and cheaper for restartable batch transcode	AI team
A5	Also request the **G/VT quota in a second region** (e.g. ap-southeast-1 Singapore or us-east-1) — GPU quota there is usually far easier than Mumbai	AI team
A6	Let S3/RDS/CloudFront accrue real spend for 1–3 weeks to build account history	—

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6. Plan B — Alternative execution if AWS rejects again

Core principle: keep storage/DB/CDN on AWS (already built and working); move only the small GPU compute. The Phase-0 GPU step reads/writes the existing `localaitv-content-mumbai` S3 bucket; at test volume the cross-cloud data cost is a few dollars — negligible. **Containerise the GPU step once** so it runs anywhere (RunPod today, AWS tomorrow) — this is the key to never being held hostage again.

Recommended GPU options (ranked for this case)

Option	What	Fit	Notes
RunPod ✅ fastest unblock	On-demand/Spot T4 / L4 / A10, per-second billing	Phase-0 **today**	No quota wall; cheap; ideal for batch transcode + ML inference. Start in hours.
Lambda Labs	Simple on-demand A10/A100	Phase-0	Clean, reliable; slightly less granular billing.
E2E Networks ✅ best India long-term	Indian GPU cloud (T4/A100), **INR billing**, Mumbai/India low latency	Phase-0 **and** a real Phase-1 option	Strong fit for a regional Indian product; easy onboarding; low latency to your users and to AWS Mumbai S3.
Paperspace / Vast.ai	Easy T4/A4000 (Paperspace) · cheapest marketplace (Vast)	Phase-0 batch	Vast = lowest cost, variable reliability — fine for non-critical test transcode.
GCP / Azure / OCI	Hyperscaler GPU	Not a fast unblock	**Same new-account GPU-quota friction as AWS** — do not expect these to be faster.

AWS-side alternatives that may dodge the wall (try in parallel)

- **Spot G/VT** (separate quota — see A4): cheaper, fine for restartable batch.
- **Different AWS region** (see A5): run the Phase-0 GPU in Singapore/US while S3 stays in Mumbai — acceptable for *validation only* (small cross-region cost; production returns to Mumbai once quota lands).
- **Mature-then-retry:** build account history (A6) and re-request — often the real unlock.

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7. Immediate / Short-term / Long-term

Immediate (this week)

- Spin up **one T4/L4 on RunPod or Lambda**; containerise the existing pipeline GPU step; run the Phase-0 G1–G4 validation against the **existing AWS S3 + RDS + CloudFront**. (Cross-cloud egress at test scale ≈ a few dollars.)
- In parallel on AWS: **Business Support upgrade (A1)**, file **Spot (A4)** and **second-region (A5)** quotas, keep the reopened On-Demand case warm.
- Resolve the still-open **DNS live-app safety** check from the previous response before any further DNS work (unrelated to quota, still outstanding).

Short-term (2–4 weeks)

- AWS spend accrues → re-request Mumbai On-Demand quota **with billing history + Business Support backing**.
- Decide **batch vs. 24/7-live** (the earlier cross-cutting question): it directly sets how much GPU quota Phase-1 needs (24/7 live ⇒ g4dn.12xlarge ⇒ ~48 vCPU — a much larger AWS ask, which makes maturing the account *now* even more important).

Long-term (scalable architecture)

- Preferred: **AWS Mumbai once the account is matured + quota granted** (co-located with S3/RDS — lowest latency/cost).

- Robust fallback: **E2E Networks (India GPU) + AWS storage** — a sound, INR-billed, low-latency hybrid for a regional product.
- Always keep the **GPU workload containerised and cloud-portable** so the scarce resource never single-vendor-locks the project.

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8. Q6 — Multi-cloud: honest advice

- **A permanent multi-cloud architecture is NOT advisable** at this stage — it adds ops complexity, cross-cloud egress cost, and team burden you don't need.
- **A temporary hybrid IS advisable** *only* as an unblock: AWS for storage/DB/CDN (already built) + a specialist/India GPU cloud for compute, **with the GPU job containerised**. Consolidate to a single home for Phase-1 once the GPU situation is settled.
- Net: hybrid **now to move fast**, single-cloud **later for simplicity** — don't institutionalise multi-cloud.

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9. Governance & credentials (unchanged)

- This is **advisory only**. Running Phase-0 validation on a non-AWS GPU **does not bypass the Phase-0 gate** — G1–G4 still must pass, the Architecture Team still re-verifies, and **only the Founder lifts the gate** before Phase-1.
- Any new provider account is created by the **Founder/CTO**; all credentials go to the **Bitwarden vault only** — never to the AI team in chat, never to the Architecture Team, never by screen-share.

Next Steps

1. **CTO:** upgrade AWS Business Support (A1) — biggest single lever.
2. **AI team:** file Spot (A4) + second-region (A5) quotas; keep the reopened case warm with softened wording (A3).
3. **AI team:** containerise the Phase-0 GPU step; stand up one GPU on RunPod/Lambda; run G1–G4 against existing AWS S3/RDS/CloudFront — **start this regardless of AWS**.
4. **Founder (internal):** decide batch vs. 24/7-live; approve Business Support spend; approve a small Plan-B GPU budget (RunPod Phase-0 ≈ tens of dollars).
5. **Architecture Team:** re-verify G1–G4 on completion (wherever the GPU ran). Gate stays held until then and until the Founder lifts it.

Sign-off

Party	Action	Status
Nagarjuna (CTO)	AWS Business Support; provider accounts → vault	☐ Pending
Sameer & Gnana	Spot + 2nd-region quota; containerise GPU step; run Phase-0 on Plan-B GPU	☐ Pending
Architecture Team	Re-verify G1–G4 on completion (any GPU host)	☐ Awaiting
Founder — Koneti Mohan Reddy	Decide batch-vs-live; approve Support + Plan-B budget; lift gate only after re-verification	◑ Holding

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Advisory only. The Phase-0 GPU need is small and should not gate the schedule — run Plan B now, push Plan A in parallel. Gate unchanged and held; credentials Bitwarden-vault-only; keep the GPU workload portable. v1.3 unchanged.